

Closed-form Second-order Data Shapley as a Spurious-Correlation Attenuator: Where it Works and Where it Doesn't

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Abstract

We adapt the closed-form In-Run Data Shapley of Wang et al. (2024) into an *in-training* per-sample reweighter for worst-group robustness. Our second-order variant augments the gradient inner product with an RMS-rescaled gradient–Hessian–gradient cross-term. We *decompose* this cross-term and, through a residual ablation, show that its benefit comes from an orthogonal, sample-specific residual rather than from mere smoothing (real > shuffle > parallel; Corrupted-CIFAR +5.9 pp, $p = 0.0015$). The mechanism reproduces across **three** from-scratch spurious regimes — color (Colored MNIST), texture (Corrupted-CIFAR), and natural images (Spurious-STL10) — where, among methods that use *no group labels at training time*, the second-order reweighter beats vanilla SGD and a corrected bi-level meta-learner (on STL10, +28.1/+17.2 pp over vanilla/Ren2018 and a +12.9 pp second-order isolation over the first-order term; all $p < 0.011$, 10 seeds). Stronger methods exist (JTT, GroupDRO); we do *not* claim state-of-the-art. On pretrained-ResNet Waterbirds fine-tuning all our variants fail to match the bi-level baseline, and we characterize this regime boundary.

Introduction

Models trained by empirical risk minimization latch onto *spurious cues* — background, color, texture — that correlate with the label in training but shift or flip at test time, so average accuracy stays high while the *worst group* collapses (Sagawa et al. 2020). The strongest remedies require group labels at training (GroupDRO) or multi-stage / bi-level pipelines (JTT, Liu et al. 2021; meta-reweighting, Ren et al. 2018).

We ask whether a *closed-form, single-pass* per-sample reweighter — no group labels in the model update, no inner optimization loop — can attenuate the spurious cue. Our starting point is In-Run Data Shapley (Wang et al. 2024), which gives a closed-form Taylor estimate of each sample’s contribution to a validation anchor *during* training. We repurpose this valuation as a training-time weight, and ask what the *second-order* curvature term adds beyond the first-order inner product.

Our contributions are:

- We repurpose closed-form In-Run Data Shapley as an in-training per-sample reweighter for worst-group robustness (Section).

- A second-order gradient–Hessian–gradient cross-term with **RMS rescaling**, which is required for stability under strong spurious bias.
- A **decomposition and residual ablation** establishing that the cross-term’s gain is a real, sample-specific directional signal — the orthogonal residual — not global smoothing (real > shuffle > parallel; sign-flip collapses).
- Evidence that the mechanism **reproduces across three distinct spurious regimes** (color, texture, natural images), with a statistically isolated second-order benefit in each.
- An **honest regime boundary**: a clean negative on pretrained-ResNet fine-tuning, with a mechanistic diagnosis and a practical recommendation.

Related Work

Data valuation. Data Shapley (Ghorbani and Zou 2019) values training points by their marginal contribution but needs many retrainings; In-Run Data Shapley (Wang et al. 2024) makes this closed-form within a single run. We repurpose that estimator — not for post-hoc valuation but as a training-time reweighter — and add a curvature term.

Reweighting for robustness and fairness. Learning to reweight (Ren et al. 2018) and FORML (Yan, Seto, and Apostoloff 2022) solve a bi-level meta problem; ARL (Lahoti et al. 2020) reweights adversarially without demographics. Our reweighter is closed-form and single-pass, with no inner optimization loop.

Worst-group robustness. GroupDRO (Sagawa et al. 2020) uses group labels; JTT (Liu et al. 2021) and IRM (Arjovsky et al. 2019) relax that requirement. These are stronger but heavier; we position Fairds as a lightweight closed-form alternative and report the gap honestly.

Shapley for fairness. FairShap (Arnaiz-Rodriguez and Oliver 2023) uses Shapley values for fair data valuation; we instead inject a *second-order* Shapley signal into the optimization trajectory itself.

Method: Fairds

Setup. At step t with parameters θ_t , a minibatch $\mathcal{B} = \{z_i\}_{i=1}^B$ and a small *group-balanced* anchor $\mathcal{D}_{\text{val}}$ (50:50,

constructed once; the model update consumes no sensitive labels). Let $g_i = \nabla_{\theta} L(z_i; \theta_t)$ and $g_{\text{val}} = \nabla_{\theta} L(\mathcal{D}_{\text{val}}; \theta_t)$.

First order (Fairds-1). $\phi_i^{(1)} = \langle g_i, g_{\text{val}} \rangle$ is the Wang et al. estimator of how much z_i reduces validation loss in one SGD step. Per-sample gradients are obtained with `torch.func.vmap` in a single autograd graph — closed form, no bi-level loop.

Second order (Fairds-2) with RMS rescaling. The second-order term adds the gradient–Hessian–gradient cross-term $c_i = \langle g_i, H_{\text{val}} g_{\text{val}} \rangle$, where $H_{\text{val}} g_{\text{val}}$ is computed by the Pearlmutter trick (Pearlmutter 1994) (one extra backward pass, no explicit Hessian). The spectrum of H_{val} varies by orders of magnitude during training, so a fixed α is unusable. We **rescale the cross-term to the same root-mean-square magnitude as the first-order term** within each minibatch:

$$\tilde{\phi}_i^{(2)} = \phi_i^{(1)} - \alpha \frac{\text{RMS}(\phi^{(1)})}{\text{RMS}(c)} c_i. \quad (1)$$

This rescaling is an algorithmic contribution: without it, second-order Fairds collapses to chance accuracy under strong spurious bias.

Pipeline. We (1) compute g_i and (for second order) $H_{\text{val}} g_{\text{val}}$; (2) update an exponential moving average EMA_i with $\tilde{\phi}_i$; (3) form per-batch weights $w_i = B \cdot \text{softmax}(\text{EMA}_i / \tau)$, clipped at quantile bounds; and (4) take a weighted SGD step $\theta_{t+1} = \theta_t - \eta \sum_i w_i \nabla_{\theta} L(z_i; \theta_t)$. For models with batch normalization we freeze BN running statistics so vmap per-sample gradients stay coupling-free.

What does the cross-term do? We find g_{val} is nearly the top eigenvector of H_{val} (alignment 0.89–0.99), so the normalized cross-term c_n is almost parallel to $\phi^{(1)}$. Decomposing $c_n = \beta \phi^{(1)} + r$ separates a *parallel* part (a curvature shrinkage, equivalent to global smoothing) from an *orthogonal residual* r . The empirical question is whether r is a real signal; we answer it in Section .

Experiments

We compare, on OOD worst-group test accuracy (the spurious cue is flipped at test time): Vanilla SGD; our Fairds-1/2 and the residual variant $\phi^{(1)} - \alpha r$; a corrected one-step bi-level meta-learner (Ren et al. 2018, a FORML proxy); JTT (Liu et al. 2021); and GroupDRO (Sagawa et al. 2020). Vanilla, Fairds, Ren2018 and JTT use *no group labels* in the update; GroupDRO uses them. Each method sees a small balanced anchor for its inner computation only.

Three spurious regimes

We use three from-scratch benchmarks with the same anchor / held-out `val_eval` / OOD-test protocol, differing only in the spurious modality: **Colored MNIST** (color; Arjovsky et al. 2019-style), **Corrupted-CIFAR** (texture: Gaussian blur vs. additive noise on car/truck), and **Spurious-STL10** (the same corruption cue on 96×96 natural images, car vs.

method	Colored MNIST (color)	Corrupted-CIFAR (texture)	Spurious-STL10 (natural)
Vanilla	0.694	0.196	0.042
Ren2018	0.796	0.375	0.151
<i>Fairds-1</i>	0.777	0.374	0.194
<i>Fairds-2</i>	0.824	0.423	0.227
<i>Fairds-2-res</i>	0.815	0.458	0.323
JTT	0.878	0.368	0.606
GroupDRO [†]	0.900	0.564	0.521

Table 1: **OOD worst-group test accuracy across three spurious regimes** (mean, 10 seeds). *Italics* = our methods. [†]uses group labels. Among no-group-label methods (top block) the second-order residual variant is best in all three regimes; JTT/GroupDRO are stronger overall.

truck). In each, the majority training group has spurious–label correlation 0.9, the minority 0.5, and the test set flips it to 0.1.

Table 1 and Figure 1 show the pattern is consistent. Among no-group-label methods, the second-order residual variant is best in every regime. On the hardest, most naturalistic regime (STL10) the margins are largest and statistically clear (seed-paired Wilcoxon, 10 seeds): +28.1 pp over vanilla ($p = 0.002$), +17.2 pp over Ren2018 ($p = 0.002$), and a **second-order isolation** of +12.9 pp over the first-order term alone ($p = 0.010$). The Colored-MNIST isolation (+4.7 pp, $p = 0.033$) and Corrupted-CIFAR isolation (+4.9 pp, $p = 0.042$) reproduce the same effect. Fairds-2 also has the lowest seed variance among no-group-label methods.

The cross-term’s signal is the orthogonal residual

To test whether the residual r is a real signal or noise, we build five arms that share $\phi^{(1)}$ but treat r differently: *phil* (first order only), *parallel* ($(1 - \alpha\beta)\phi^{(1)}$, the shrinkage part), *residual-real* ($\phi^{(1)} - \alpha r$), *residual-shuffle* (r permuted within class× $\phi^{(1)}$ -quantile bins), and *sign-flip* ($\phi^{(1)} + \alpha r$). Figure 2 reports the result.

The ordering is unambiguous: **real** > **shuffle** > **parallel**, and sign-flip collapses. On Corrupted-CIFAR (20 seeds), residual-real beats residual-shuffle by +5.9 pp ($p = 0.0015$) and the parallel/shrinkage arm by +9.2 pp ($p < 10^{-4}$); sign-flipping the residual destroys accuracy (0.07, $p = 5 \times 10^{-4}$). If r were noise, shuffling it within matched bins would not matter — but it does. Thus the second-order benefit is a real, sample-specific *directional* signal, not global smoothing.

Controlled mechanism probe

On a controlled spurious-feature 2-layer MLP (a single strongly label-correlated feature in the majority group), the per-sample ϕ_i of Fairds-2 assign *lower* value to majority than minority samples (Mann–Whitney $U(\text{maj} < \text{min})$, $p = 1.89 \times 10^{-8}$ at ratio 0.9), with $\Delta\phi(\text{maj} - \text{min})$ monotonically decreasing as imbalance grows. This is the predicted mechanism: the cross-term penalizes majority gradi-

Fairds-2 (residual) beats no-group-label baselines across 3 spurious regimes (grey/blue = no group label; red = uses group label or 2-stage)

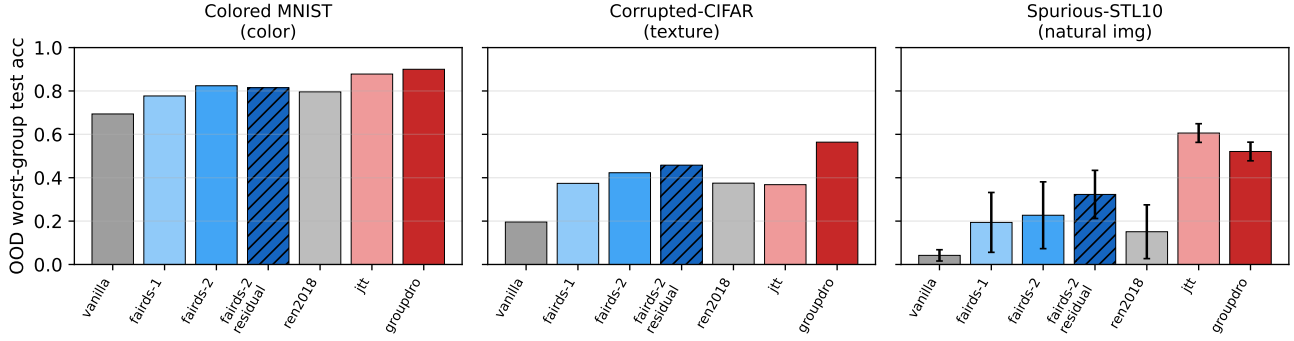


Figure 1: **The second-order mechanism reproduces across three spurious regimes.** OOD worst-group test accuracy (higher is better). Grey/blue bars use *no* group label at training; red bars use a group label or a two-stage procedure. Fairds-2 (residual; hatched) beats every no-group-label baseline in all three regimes. Error bars (STL10) are ± 1 std over 10 seeds.

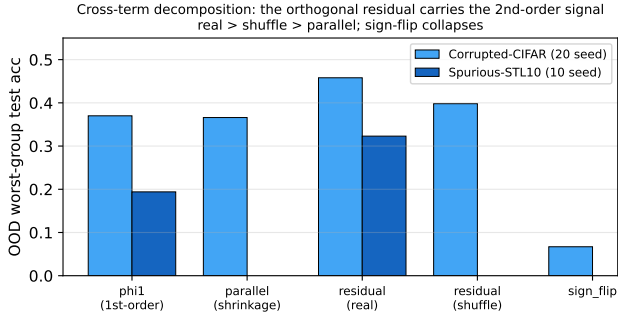


Figure 2: **Residual ablation.** Decomposing the cross-term, the orthogonal residual carries the signal: real > shuffle > parallel; sign-flip collapses. STL10 measures *phi1* and *residual-real* at 10 seeds.

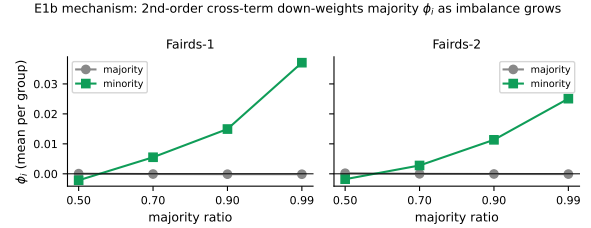


Figure 3: **Controlled probe.** Mean per-sample ϕ_i by group vs. imbalance ratio on the spurious-feature MLP. Both Fairds variants down-weight majority ϕ_i ; the second-order cross-term amplifies the effect.

Discussion and Conclusion

ents aligned with the validation Hessian’s dominant directions.

Honest scope

Stronger methods exist. JTT and GroupDRO reach higher worst-group accuracy in most regimes (Table 1). Fairds-2’s contribution is the closed-form, single-pass cross-term mechanism, not peak accuracy.

A regime where it fails. On the canonical Waterbirds benchmark (Sagawa et al. 2020) with a *pretrained* ResNet-18, three Fairds variants (default EMA; warmup; per-step no-EMA) all fail to match the corrected bi-level baseline (Ren2018 test_worst ≈ 0.49 vs. Fairds-2 ≤ 0.20 , $p < 0.005$). With a strong pretrained extractor, early per-sample gradients are small and noisy, so the EMA buffer accumulates noise and reweighting destabilizes training. When the backbone is frozen and Shapley acts only on a freshly initialized last layer (a DFR-style setup), Fairds recovers to parity with the bi-level baseline and with last-layer retraining — but this is a boundary result, not a win.

Closed-form data valuation can shape training dynamics, but *within a characterized regime*. The second-order cross-term helps because of its orthogonal residual — a real, per-sample directional signal that down-weights majority gradients aligned with the validation curvature — and this mechanism reproduces across color, texture, and natural-image spurious cues from scratch. It does not transfer to pre-trained fine-tuning, where bi-level meta-learning is the better tool; the failure has a clear mechanistic cause (noisy small gradients corrupting the EMA buffer). The method is not state-of-the-art: two-stage (JTT) and group-label (GroupDRO) methods remain stronger, and the second-order variant costs roughly $7.7\times$ vanilla wall-clock (the first-order variant, $4.3\times$, is cheaper than the bi-level baseline). Its value is conceptual: a single-pass, closed-form valuation whose curvature term provably carries a usable robustness signal. Characterizing why the residual is informative in some architectures but not others, and reducing the second-order overhead, are the natural next steps.

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